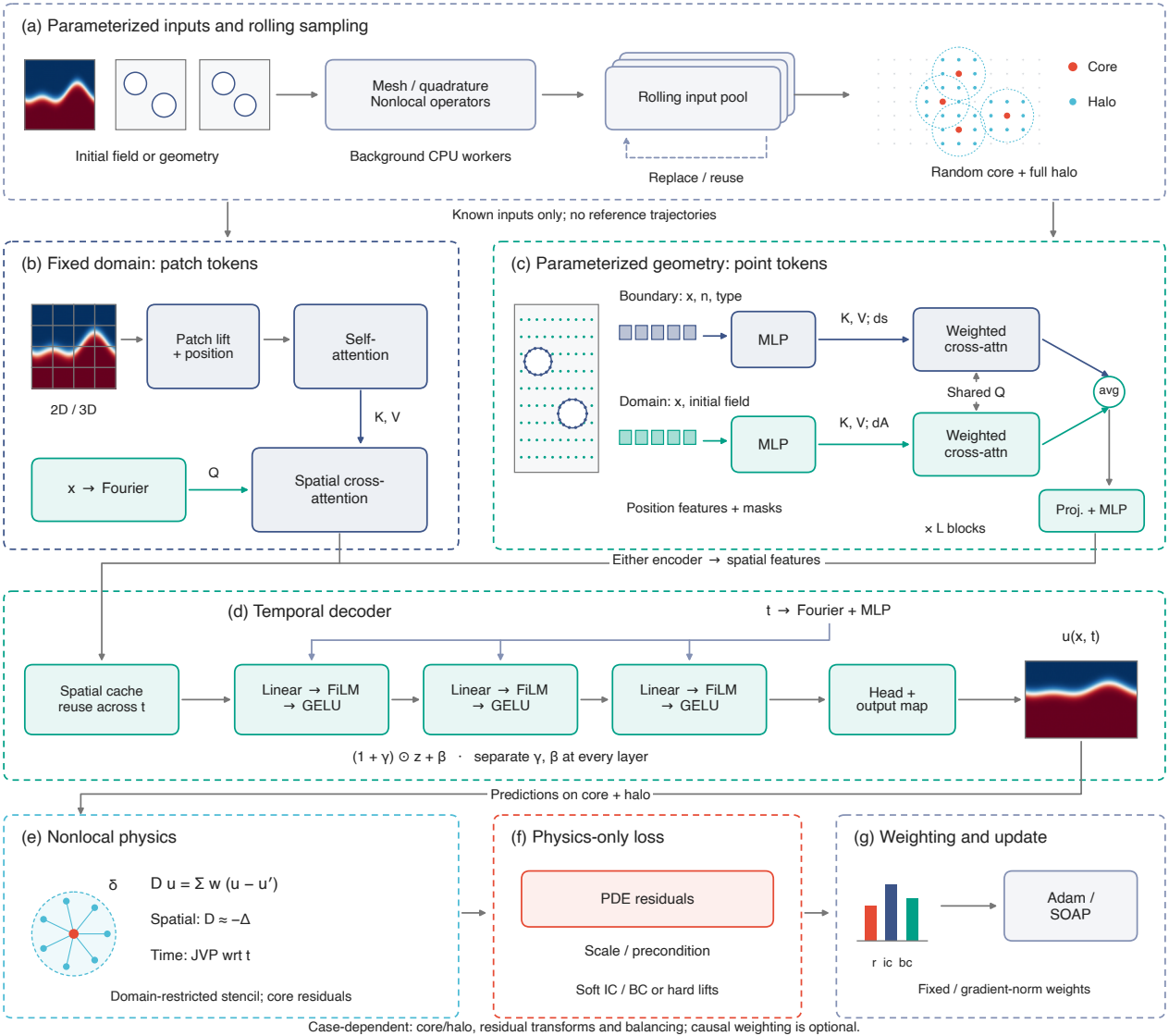

(a) Parameterized inputs and rolling sampling
Initial field or geometry
Mesh / quadrature
Nonlocal operators
Background CPU workers
Rolling input pool
Replace / reuse
Core
Halo
Random core + full halo
Known inputs only; no reference trajectories
(b) Fixed domain: patch tokens
2D / 3D
Patch lift
+ position
Self-
attention
K, V
x → Fourier
Q
Spatial cross-
attention
(c) Parameterized geometry: point tokens
Boundary: x, n, type
MLP
K, V; ds
Weighted
cross-attn
Shared Q
avg
Domain: x, initial field
MLP
K, V; dA
Weighted
cross-attn
Position features + masks
× L blocks
Proj. + MLP
Either encoder → spatial features
(d) Temporal decoder
t → Fourier + MLP
u(x, t)
Spatial cache
reuse across t
Linear → FiLM
→ GELU
Linear → FiLM
→ GELU
Linear → FiLM
→ GELU
Head +
output map
(1 + γ) ⊙ z + β · separate γ, β at every layer
Predictions on core + halo
(e) Nonlocal physics
δ
D u = Σ w (u – u′)
Spatial: D ≈ –Δ
Time: JVP wrt t
Domain-restricted stencil; core residuals
(f) Physics-only loss
PDE residuals
Scale / precondition
Soft IC / BC or hard lifts
(g) Weighting and update
r ic bc
Adam /
SOAP
Fixed / gradient-norm weights
Case-dependent: core/halo, residual transforms and balancing; causal weighting is optional.